TNRC6B-Related Neurodevelopmental Disorder — Comprehensive Disease Report

Disease: TNRC6B-Related Neurodevelopmental Disorder **Synonym (OMIM):** Global developmental delay with speech and behavioral abnormalities (GDSBA) **OMIM:** 619243 (phenotype) · 610740 (gene) · **MONDO:** MONDO:0030995 · **Gene:** *TNRC6B* (HGNC:28368; NCBI Gene 23112; 22q13.1) **Category:** Mendelian, autosomal dominant

Summary

TNRC6B-related neurodevelopmental disorder (NDD) is an **ultra-rare, autosomal dominant Mendelian condition** caused by heterozygous, predominantly loss-of-function variants in *TNRC6B*, a gene encoding a GW182-family scaffold protein essential for microRNA (miRNA)-mediated gene silencing. The disorder was first systematically delineated in 2020 in a cohort of 17 patients (PMID: 32152250), in which developmental delay/intellectual disability (DD/ID) was present in 100% of subjects, accompanied by speech delay (94%), motor delay (71–82%), autism or autistic traits (76%), ADHD (65%), and variable musculoskeletal and dysmorphic features. It carries the OMIM designation **619243 "Global developmental delay with speech and behavioral abnormalities"** and is formally described as an autosomal dominant disease caused by *TNRC6B* variants (PMID: 39115759).

Mechanistically, *TNRC6B* encodes one of three human GW182 paralogs (TNRC6A/B/C). The protein scaffolds the miRNA-induced silencing complex (miRISC): its glycine-tryptophan (GW/WG)-rich N-terminus binds Argonaute (AGO2), while its C-terminal silencing domain recruits poly(A)-binding protein (PABP) and the CCR4-NOT deadenylation machinery to execute target-mRNA repression. Pathogenic variants are heterozygous and reduce functional dosage (haploinsufficiency); the gene is highly loss-of-function intolerant in population databases (pLI ≈ 1.0). Because the same miRNA machinery (AGO2/TNRC6B/FMRP/MOV10) governs neural stem-cell differentiation, dendritic maturation, and DICER-dependent miRNA production,

reduced TNRC6B dosage is proposed to dysregulate the post-transcriptional programs required for normal neuronal differentiation and cortical development, producing the neurodevelopmental phenotype.

Clinically, the disorder presents congenitally or in early childhood with a **static (non-progressive)** picture. Diagnosis is molecular — via trio whole-exome (WES) or whole-genome sequencing (WGS), supplemented by chromosomal microarray for 22q13.1 deletions, and by minigene/RNA assays to resolve splice-altering or synonymous variants of uncertain significance. There is **no disease-specific or targeted therapy**; management is entirely supportive (early intervention; speech, physical, and occupational therapy; special education; and symptomatic treatment of seizures, ADHD, and behavioral comorbidities). Life expectancy is expected to be normal or near-normal. The remainder of this report expands each disease-characteristic section with the available evidence and flags where information is absent from the primary literature.

Key Findings

Finding 1 — TNRC6B-related NDD is an autosomal dominant disorder caused by heterozygous *TNRC6B* variants (OMIM 619243)

The foundational clinical delineation comes from Granadillo et al. (2020), who characterized **17 patients** with heterozygous *TNRC6B* variants. The core phenotype was DD/ID in **17/17 (100%)**, with speech delay in **16/17 (94%)**, fine motor delay in **14/17 (82%)**, gross motor delay in **12/17 (71%)**, autism or autistic traits in **13/17 (76%)**, ADHD in **11/17 (65%)**, other behavioral problems in **7/17**, and musculoskeletal findings in **12/17 (71%)**. As stated in the abstract: *"Clinical findings included DD/ID (17/17) (speech delay in 94% (16/17), fine motor delay in 82% (14/17) and gross motor delay in 71% (12/17) of subjects), autism or autistic traits (13/17), attention deficit and hyperactivity disorder (ADHD) (11/17), other behavioural problems (7/17) and musculoskeletal findings (12/17)"* (PMID: 32152250). The disorder carries OMIM 619243 and is described as *"an autosomal dominant disease caused by variants in TNRC6B gene"* (PMID: 39115759). This establishes both the inheritance pattern and the quantitative phenotype frequencies used throughout this report.

Finding 2 — TNRC6B encodes a GW182-family scaffold essential for miRNA-mediated gene silencing

TNRC6B is one of three human GW182 paralogs (TNRC6A/B/C). Its GW/WG-rich N-terminus contains multiple Argonaute (AGO2)-binding sites that engage tryptophan-binding pockets in the AGO2 PIWI domain, while the C-terminal RRM/silencing domain recruits PABP and the CCR4-NOT deadenylation machinery. Functionally, TNRC6B is a core component of miRISC: *"two core protein components of human miRISC, Argonaute2 (Ago2) and TNRC6B, condense into phase-separated droplets in vitro and in live cells"* (PMID: 29576456), and *"the multivalent binding of TNRC6 enables cooperative binding of miRNA-AGO complexes to target RNAs, thereby explaining the basis of cooperative action"* (PMID: 32661162). GW182/TNRC6 proteins are required for miRNA activity: *"GW182 is required for transcriptional activation and for the activity of miRNAs"* (PMID: 30086238). These data define the molecular role that is compromised in disease.

Finding 3 — Pathogenic variants are predominantly loss-of-function, including a pathogenic synonymous splicing variant

Reported disease-causing variants span the full loss-of-function spectrum: nonsense, frameshift (e.g., c.1632delC p.Leu546fs*63), missense (e.g., c.335C>T p.Pro112Leu; c.1409A>G p.Asp470Ser), splice-altering, and whole/partial-gene deletions at 22q13.1. Notably, a synonymous variant c.3141G>A was proven pathogenic by minigene assay through induction of exon 7 skipping:* "we demonstrated that c.3141G > A variant induced exon 7 skipping and the synonymous variant was pathogenic" *(PMID: 39115759). Yang et al. reported* "novel de novo likely pathogenic or pathogenic TNRC6B variants c.335C>T (p.Pro112Leu) and c.1632delC (p.Leu546fs*63)"* (PMID: 38404251). The gene is highly loss-of-function intolerant in gnomAD (pLI ≈ 1.0), consistent with **haploinsufficiency** as the disease mechanism.

Finding 4 — Expanded phenotype includes dysmorphism, growth abnormalities, café-au-lait spots, seizures, and metabolic abnormalities

Beyond the core NDD features, the phenotypic spectrum has broadened. Yang et al. (2024) reported *"DD/ID, delayed speech, ADHD, behavioral abnormalities, short stature, low body weight, café-au-lait spots, metabolic abnormalities, and facial dysmorphism including coarse facial features, sparse hair, frontal bossing, hypertelorism, amblyopia, strabismus, and downslanted palpebral fissures"* (PMID: 38404251). Importantly, there is no recognizable facial

gestalt: *"The majority of patients exhibited some dysmorphic features but no recognisable gestalt was identified"* (PMID: 32152250). Epilepsy co-occurring with ASD has been reported as a presenting feature (PMID: 37348364).

Finding 5 — The miRNA/miRISC machinery regulates neuronal differentiation and dendrite development, linking TNRC6B loss to neurodevelopmental phenotypes

TNRC6B operates within the neural miRNA machinery in a developmental-stage-specific manner: *"Developmental regulation of the processing pathway correlates with differential localization of the proteins Argonaute, FMRP, MOV10, and TNRC6B in self-renewing stem cells and neurons"* (PMID: 17167072). The same AGO2-cofactor module controls DICER and dendrite formation: *"the MOV10-FMRP-AGO2 complex regulates DICER expression, revealing a novel mechanism for regulation of miRNA production required for normal neuronal morphology"* (PMID: 34847178). This provides the mechanistic bridge between TNRC6B haploinsufficiency and impaired neurodevelopment.

Finding 6 — Mouse Tnrc6 knockouts show paralog-specific essentiality, but no established Tnrc6b neurodevelopmental model exists

Guo et al. (2017) generated Tnrc6a/b/c knockout mice: *"TNRC6a mutants die at mid-gestation, while b- and c- deleted mice are born at a Mendelian ratio. However, the majority of TNRC6b and all TNRC6c mutants die within 24h after birth"* (PMID: 28811219). Homozygous loss is largely perinatal-lethal and the paralogs are not fully redundant in vivo. The mouse ortholog is *Tnrc6b* (NCBI Gene 213988); the human gene is *TNRC6B* (NCBI Gene 23112, HGNC:28368). No heterozygous mouse model recapitulating the human neurodevelopmental phenotype has yet been reported.

Finding 7 — Diagnosis relies on exome/genome sequencing; management is supportive with no disease-specific therapy

TNRC6B-related NDD is a molecular diagnosis made by WES/WGS and chromosomal microarray (for 22q13.1 deletions). Professional guidance supports broad genetic testing: *"professional medical organizations recommend genetic testing for all individuals with a NDD"* (PMID: 41274961). Functional assays resolve ambiguous variants: *"WES combined with minigene assay establishes a robust basis for genetic counseling and diagnosing diseases"* (PMID: 39115759). No approved disease-specific therapy exists; management is supportive.

Finding 8 — Autosomal dominant inheritance with both de novo and inherited variants, variable expressivity, and ultra-rare prevalence

Most pathogenic variants arise de novo (*"novel de novo likely pathogenic or pathogenic TNRC6B variants"* — PMID: 38404251), confirming autosomal dominant inheritance (*"autosomal dominant disease caused by variants in TNRC6B gene"* — PMID: 39115759). However, inherited transmission occurs: a three-generation family segregating a 22q13.1 deletion involving *TNRC6B* has been reported (PMID: 42074582), demonstrating vertical transmission with variable expressivity and probable incomplete penetrance. Fewer than ~50–100 molecularly confirmed cases exist; no formal prevalence/incidence has been established.

Finding 9 — The central nervous system is the primary affected organ, with multisystem secondary involvement

The brain/CNS is the primary target (DD/ID, speech/language delay, ASD, ADHD, seizures). Secondary systems include musculoskeletal (12/17 in Granadillo 2020), craniofacial/skeletal, ophthalmologic (strabismus, amblyopia), growth/endocrine-metabolic, and integumentary (café-au-lait macules) systems — *"autism or autistic traits (13/17), attention deficit and hyperactivity disorder (ADHD) (11/17), other behavioural problems (7/17) and musculoskeletal findings (12/17)"* (PMID: 32152250).

Section-by-Section Report

1. Disease Information

TNRC6B-related neurodevelopmental disorder is a **monogenic, autosomal dominant NDD** caused by heterozygous variants in *TNRC6B*. It is characterized by developmental delay / intellectual disability, prominent speech and language delay, autism spectrum features, ADHD, and variable dysmorphic, growth, and behavioral abnormalities.

Key identifiers: - **OMIM:** 619243 ("Global developmental delay with speech and behavioral abnormalities"; phenotype) / 610740 (*TNRC6B* gene) - **MONDO:** MONDO:0030995 - **Gene:** *TNRC6B* — HGNC:28368; NCBI Gene 23112; chromosome 22q13.1 - **Orphanet / ICD-10 / ICD-11 / MeSH:** No dedicated disease-specific code identified in the primary literature reviewed; the condition is captured under broad NDD/intellectual-disability categories.

Synonyms / alternative names: Global developmental delay with speech and behavioral abnormalities (GDSBA); TNRC6B deficiency syndrome; TNRC6B-related developmental delay.

Information source type: The disease-level knowledge is **aggregated from individual-patient case series and case reports** (notably the 17-patient Granadillo cohort and subsequent single/multi-patient reports), not from EHR-scale or registry-level datasets.

2. Etiology

Causal factors: Purely genetic — heterozygous, predominantly loss-of-function variants in *TNRC6B*. There is no evidence for environmental, infectious, or multifactorial causation.

Genetic risk factors: The causal variants themselves (nonsense, frameshift, splice-site, missense, synonymous splice-altering, and 22q13.1 deletions). No susceptibility loci or validated modifier genes have been established. In one case, co-occurrence of a *TNRC6B* variant with an Xq28/*MECP2* microdeletion was associated with a more severe phenotype, suggesting that a second genomic lesion can act as a modifier (PMID: 41147347).

Environmental risk factors / protective factors / gene-environment interactions: None identified. As a Mendelian dominant disorder, environmental and lifestyle factors are not established contributors, and no protective alleles or gene-environment interactions are reported. *Not applicable / not available for this disease.*

3. Phenotypes

Phenotype	Type	Frequency	Onset	Course	Suggested HPO
Developmental delay / intellectual disability	Cognitive/ behavioral	17/17 (100%)	Infancy/ early childhood	Static	HP:0001263 / HP:0001249
Speech / language delay	Cognitive/ behavioral	16/17 (94%)	Early childhood	Static	HP:0000750
Fine motor delay	Motor	14/17 (82%)	Infancy	Static	HP:0010862
Gross motor delay	Motor	12/17 (71%)	Infancy	Static	HP:0002194
Autism / autistic traits	Behavioral	13/17 (76%)	Early childhood	Static	HP:0000717
ADHD	Behavioral	11/17 (65%)	Childhood	Static	HP:0007018
Other behavioral problems	Behavioral	7/17 (41%)	Childhood	Variable	HP:0000708
Musculoskeletal findings	Physical	12/17 (71%)	Congenital/ childhood	Static	HP:0033127
Facial dysmorphism (no gestalt)	Physical	Majority	Congenital	Static	HP:0001999
Short stature / low body weight	Growth	Variable	Childhood	Variable	HP:0004322 / HP:0004325
Microcephaly	Physical	Occasional	Congenital/ childhood	Static	HP:0000252
Café-au-lait macules	Physical	Occasional	Childhood	Stable	HP:0000957
Strabismus / amblyopia	Ophthalmologic	Occasional	Childhood	Variable	HP:0000486 / HP:0000646
Seizures / epilepsy	Neurological	Occasional	Childhood		HP:0001250

Phenotype	Type	Frequency	Onset	Course	Suggested HPO
				Variable/ episodic	
Metabolic abnormalities	Laboratory	Occasional	Childhood	Variable	HP:0001939

Severity and quality-of-life impact: Severity is variable but generally moderate; the dominant burden is cognitive, communicative, and behavioral, substantially affecting education, communication, and independent daily functioning. Formal QoL instrument data (EQ-5D, SF-36, PROMIS) are *not available* for this disorder.

4. Genetic/Molecular Information

Causal gene: *TNRC6B* (trinucleotide repeat containing 6B), 22q13.1; HGNC:28368; NCBI Gene 23112; OMIM gene 610740; disorder OMIM 619243.

Pathogenic variants (representative):

Variant (cDNA)	Protein	Type	Classification	Origin
c.3141G>A	(synonymous) → exon 7 skipping	Splice-altering synonymous	Pathogenic (minigene-proven)	—
c.335C>T	p.Pro112Leu	Missense	Likely pathogenic/ pathogenic	De novo
c.1632delC	p.Leu546fs*63	Frameshift	Pathogenic	De novo
c.1409A>G	p.Asp470Ser	Missense	VUS/likely pathogenic	—
22q13.1 deletion	whole/partial gene loss	Structural/CNV	Pathogenic	Inherited (3-gen family)

Allele frequency: Pathogenic variants are absent or ultra-rare in gnomAD; *TNRC6B* is highly loss-of-function intolerant (pLI ≈ 1.0). **Origin:** Germline (de novo most common; inherited in deletion families). **Functional consequence:** Loss of function / haploinsufficiency.

Modifier genes: No validated modifiers; a second-site *MECP2* deletion worsened phenotype in one case (PMID: 41147347). **Epigenetic information:** No disorder-specific DNA-methylation signature ("episignature") has been reported — *knowledge gap*. **Chromosomal abnormalities:** 22q13.1 deletions involving *TNRC6B* (PMID: 42074582).

5. Environmental Information

No environmental, lifestyle, or infectious contributors are implicated. This is a monogenic disorder. *Not applicable.*

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. A heterozygous loss-of-function *TNRC6B* variant (nonsense/frameshift/splice/missense/deletion) **leads to** reduced production of functional TNRC6B protein (haploinsufficiency; demonstrated by LoF-intolerance and minigene-proven exon skipping).
2. Reduced TNRC6B **results in** diminished GW182-scaffold availability for assembly of the miRNA-induced silencing complex (miRISC) — TNRC6B normally bridges AGO2 to PABP and the CCR4-NOT deadenylase (demonstrated biochemically).
3. Impaired miRISC scaffolding **leads to** less efficient cooperative/multivalent binding of miRNA-AGO complexes to target mRNAs and reduced phase-separated condensate formation (demonstrated in vitro/in cells; PMID: 29576456, PMID: 32661162).
4. Weakened miRNA-mediated silencing **results in** derepression of miRNA target transcripts and broadly dysregulated post-transcriptional gene expression (inferred from loss of miRNA activity; PMID: 30086238).
5. Because this machinery (AGO2/TNRC6B/FMRP/MOV10) is developmentally deployed in neural stem cells and neurons and regulates DICER-dependent miRNA production, its disruption **leads to** impaired neural differentiation and dendritic maturation (demonstrated for the AGO2-cofactor module; PMID: 17167072, PMID: 34847178).
6. Abnormal neuronal differentiation and connectivity during critical developmental windows **results in** the clinical neurodevelopmental phenotype — DD/ID, speech delay, ASD, ADHD, and seizures (inferred; the cell-biological-to-clinical link is not directly demonstrated in patients).

Branch: Beyond the CNS, TNRC6B-dependent miRNA regulation operates in multiple tissues, plausibly explaining the secondary musculoskeletal, craniofacial, growth, skin (café-au-lait), and metabolic features (inferred, pleiotropic).

```
TNRC6B LoF variant
      │ (haploinsufficiency)
      ▼
↓ functional TNRC6B (GW182 scaffold)
      │
      ▼
Defective miRISC assembly (AGO2–TNRC6B–PABP–CCR4/NOT)
      │
      ▼
↓ miRNA-mediated silencing / condensate formation
      │
      ├──► CNS branch: impaired neural differentiation & dendrite maturation ──► DD/ID, speech d
      └──► Systemic branch: pleiotropic miRNA dysregulation ──► dysmorphism, growth/metabolic, s
```

Molecular pathway: miRNA-mediated post-transcriptional gene silencing (RNA interference). **Cellular processes:** neural stem-cell differentiation, dendritic/neuronal morphogenesis, mRNA deadenylation/decay. **Protein dysfunction:** loss of function of a scaffolding protein. **Subcellular localization:** cytoplasmic P-bodies / ribonucleoprotein granules (GO:0000932 P-body; GO:0036464 cytoplasmic ribonucleoprotein granule). **Suggested GO BP terms:** GO:0035195 (miRNA-mediated gene silencing), GO:0031047 (regulation of gene silencing by RNA), GO:0021954 (CNS neuron development), GO:0016358 (dendrite development). **Suggested CL terms:** CL:0000540 (neuron), CL:0000047 (neuronal stem cell). No disorder-specific transcriptomic, proteomic, metabolomic, or CRISPR-screen data in patients are available — *knowledge gap*.

7. Anatomical Structures Affected

- **Primary organ:** Brain / central nervous system — UBERON:0000955 (brain), UBERON:0001017 (CNS), UBERON:0000956 (cerebral cortex).
- **Secondary involvement:** Musculoskeletal system (UBERON:0002204), craniofacial skeleton, eye (UBERON:0000970), skin (UBERON:0002097; café-au-lait macules), and growth/endocrine-metabolic axes.
- **Tissue/cell level:** Nervous tissue; neurons (CL:0000540) and neural progenitors (CL:0000047).
- **Subcellular:** Cytoplasmic P-bodies / RNP granules (GO:0000932, GO:0036464).
- **Localization/lateralization:** CNS involvement is bilateral/diffuse; no consistent lateralization.

8. Temporal Development

- **Onset:** Congenital to early childhood; delays typically recognized in infancy/toddlerhood.
- **Onset pattern:** Insidious/chronic (developmental).
- **Progression: Static (non-progressive)** neurodevelopmental course; behavioral problems may fluctuate, and seizures are episodic when present.
- **Duration:** Chronic, lifelong.
- **Remission:** No spontaneous remission; developmental gains occur with intervention.
- **Critical period:** Early childhood is the key window for developmental/therapeutic intervention.

9. Inheritance and Population

- **Inheritance:** Autosomal dominant (PMID: 39115759).
- **Variant origin:** Predominantly de novo (PMID: 38404251); inherited transmission documented in a three-generation 22q13.1-deletion family (PMID: 42074582).
- **Penetrance/expressivity:** Variable expressivity; likely incomplete penetrance (inferred from familial segregation). Precise penetrance not quantified.
- **Anticipation / germline mosaicism / founder effects / consanguinity:** Not established (not a repeat-expansion disorder; dominant mechanism does not require consanguinity).
- **Epidemiology:** Ultra-rare; fewer than ~50–100 molecularly confirmed cases reported. No formal prevalence or incidence; no Orphanet prevalence estimate.
- **Population/sex distribution:** Reported across diverse populations (European, Chinese, others). No established founder effect or strong sex bias.

10. Diagnostics

- **Primary approach:** Molecular genetic testing. Trio **WES/WGS** is the first-tier test for NDD and identifies de novo *TNRC6B* variants; **chromosomal microarray (CMA)** detects 22q13.1 deletions. *"Professional medical organizations recommend genetic testing for all individuals with a NDD"* (PMID: 41274961).
- **Variant interpretation:** Minigene / RNA splicing assays resolve synonymous and splice-region VUS — *"WES combined with minigene assay establishes a robust basis for genetic counseling and diagnosing diseases"* (PMID: 39115759).
- **Gene panels:** Broad intellectual-disability/autism/epilepsy NGS panels that include *TNRC6B*.

- **Imaging / EEG:** Brain MRI and EEG are used for phenotyping (seizures), not diagnostic of the gene.
- **Biomarkers:** No specific biochemical biomarker. **Karyotyping/FISH/mtDNA/repeat testing:** generally not indicated (CMA preferred for the deletion subtype).
- **Clinical criteria:** No standalone diagnostic criteria; diagnosis = compatible NDD phenotype + pathogenic *TNRC6B* variant.
- **Differential diagnosis:** Other monogenic NDD/ASD genes, 22q13 (Phelan-McDermid/*SHANK3*) region disorders, Rett syndrome (*MECP2*), and syndromic ID with café-au-lait (e.g., neurofibromatosis type 1, RASopathies, constitutional mismatch-repair deficiency) when pigmentary findings are present.
- **Screening:** No newborn/population screening; cascade testing for at-risk relatives in inherited cases.

11. Outcome/Prognosis

- **Survival/mortality:** Life expectancy is expected to be **normal or near-normal**; the disorder is not known to be life-limiting. No disease-specific mortality data.
- **Morbidity/disability:** Lifelong cognitive, communicative, and behavioral disability is the main burden; degree varies.
- **Complications:** Seizures, behavioral/psychiatric comorbidities (ASD, ADHD), feeding/growth issues in a subset.
- **Recovery potential:** Developmental gains achievable with early intervention; underlying deficits are static and persistent.
- **Prognostic factors:** Severity of baseline ID, presence of epilepsy, and co-occurring genomic lesions (e.g., concurrent *MECP2* deletion associated with more severe delay and malnutrition — PMID: 41147347).

12. Treatment

- **Disease-specific/targeted therapy: None approved or available.** Management is supportive.
- **Pharmacotherapy (symptomatic):** Antiepileptic drugs for seizures; stimulant/non-stimulant ADHD medications; agents for behavioral/psychiatric comorbidities as clinically indicated (NCIT: anticonvulsant therapy; CNS stimulant therapy).
- **Rehabilitative/supportive:** Early intervention programs; speech-language therapy; physical and occupational therapy; special education; behavioral therapy for ASD;

nutritional/feeding support where needed (NCIT: speech therapy; physical therapy; occupational therapy; supportive care).

- **Advanced/experimental therapeutics:** Gene-directed and ASO approaches for rare NDDs remain investigational; none specific to *TNRC6B*. No NCT-registered disorder-specific trials identified.
- **Pharmacogenomics / personalized medicine:** Not established for this disorder.
- **Treatment strategy:** Individualized, multidisciplinary, symptom-directed care.

13. Prevention

- **Primary prevention:** Not possible for de novo variants. **Genetic counseling** for families, especially where a variant is inherited (e.g., 22q13.1-deletion kindreds), informs recurrence risk (up to 50% for a transmitting dominant carrier).
- **Secondary prevention:** Early developmental surveillance and intervention to optimize outcomes.
- **Tertiary prevention:** Proactive management of seizures, behavioral comorbidities, and growth/nutrition to limit complications.
- **Reproductive options:** Prenatal diagnosis and preimplantation genetic testing are available for families with a known pathogenic variant.
- **Immunization / public-health / environmental measures:** Not applicable.

14. Other Species / Natural Disease

- **Orthologs:** Mouse *Tnrc6b* (NCBI Gene 213988) is orthologous to human *TNRC6B* (NCBI Gene 23112). The GW182/TNRC6 family is evolutionarily conserved across metazoans.
- **Natural disease / veterinary relevance:** No naturally occurring *TNRC6B* disease reported in companion animals or wildlife (OMIA search negative). *Not applicable.*
- **Zoonotic potential / cross-species transmission:** Not applicable (genetic disorder).

15. Model Organisms

- **Mouse (Mus musculus, NCBI Taxon 10090):** Constitutive knockouts of *Tnrc6a/b/c* established paralog-specific essentiality. *"TNRC6a mutants die at mid-gestation, while b- and c- deleted mice are born at a Mendelian ratio. However, the majority of TNRC6b and all TNRC6c mutants die within 24h after birth"* (PMID: 28811219). Homozygous *Tnrc6b* loss is largely perinatal-lethal.

- **Phenotype recapitulation:** Current knockouts model gene essentiality and miRNA biology but **do not model the human heterozygous neurodevelopmental phenotype**; no heterozygous or conditional CNS *Tnrc6b* model of the NDD has been reported.
- **Other systems:** In vitro biochemical and cell-based systems (HEK293/HeLa; tethering and phase-separation assays) have defined TNRC6B's silencing and condensate functions (PMID: 29576456, PMID: 19470757). Patient-derived iPSC/neuronal or organoid models are a logical but currently unreported resource.

Mechanistic Model / Interpretation

The unifying model is **dosage-sensitive disruption of microRNA-mediated post-transcriptional gene regulation during neurodevelopment.** TNRC6B is a GW182-family scaffold that is one of the three structural pillars of miRISC. Its N-terminal GW/WG repeats tether multiple Argonaute molecules, enabling cooperative, multivalent engagement of miRNA-AGO complexes with target mRNAs and driving liquid-liquid phase separation into P-body-like condensates; its C-terminal silencing domain couples the complex to PABP and the CCR4-NOT deadenylase to effect repression. Losing one functional allele reduces scaffold concentration below the threshold required for robust silencing. Because neural stem-cell specification, neuronal differentiation, DICER-dependent miRNA production, and dendritic maturation all depend on this AGO2/TNRC6B/FMRP/MOV10 module in a stage-specific manner, haploinsufficiency plausibly perturbs the fine post-transcriptional tuning that cortical and neuronal development require — yielding the clinical NDD. The multisystem (craniofacial, skeletal, growth, skin, metabolic) features reflect the pleiotropy of miRNA regulation across tissues.

Two caveats temper this model. First, the patient-level link (steps 5→6 in the causal chain) is **inferred**: no patient neurons, transcriptomes, or methylation signatures have been profiled to show derepressed miRNA targets in vivo. Second, mouse genetics show the paralogs are **not redundant** and that homozygous loss is lethal, so the heterozygous, viable human disorder occupies a dosage window not yet modeled experimentally.

Evidence Base

PMID	Title (abbrev.)	Role in this report
32152250	*Pathogenic variants in TNRC6B…* (Granadillo 2020)	Foundational 17-patient cohort; core phenotype frequencies
39115759	*Synonymous variant c.3141G>A altering splicing*	AD inheritance, OMIM 619243, LoF via exon skipping, diagnostic approach
38404251	*Novel variants…café-au-lait, metabolic* (Yang 2024)	Expanded phenotype; de novo missense/ frameshift variants
29576456	*Phase transitions in human miRISC*	TNRC6B as core miRISC condensate component
32661162	*Biochemical basis for cooperative miRNA action*	Multivalent TNRC6B scaffolding
30086238	*GW182 scaffolding requirement*	GW182/TNRC6 required for miRNA activity
17167072	*let-7 during neural specification*	TNRC6B in neural differentiation machinery
34847178	*FMRP/MOV10 regulate Dicer1 & dendrites*	AGO2-module → dendrite development link
28811219	*Tnrc6c essential in lung; Tnrc6 KO mice*	Paralog-specific mouse lethality
41274961	*Precision diagnostics in rare NDDs*	Genetic testing recommended for all NDD
42074582	*3-generation 22q13.1 deletion family*	Inherited transmission; variable expressivity
37348364	*Epilepsy and ASD from TNRC6B variant*	Seizure/ASD presentation
41147347	*TNRC6B + Xq28/MECP2 microdeletion*	Modifier effect; more severe phenotype
19470757	*Functional dissection of human TNRC6*	Domain architecture; RRM/silencing domain
19398495	*Mammalian GW182 Ago-binding sites*	Multiple AGO2-binding sites in N-terminus
38300321	*WES in severe DLD (Swedish cohort)*	TNRC6B among diagnostic yield in language disorder
42607197	*Craniosynostosis genetics*	TNRC6B among NDD-associated variants

Challenging/qualifying evidence: PMID: 25228304 (TCF20/TNRC6B at a chr22 breakpoint) found **no significant association of TNRC6B mutations with ASD** in a screened family set, attributing the phenotype to TCF20 — an early signal that individual TNRC6B variants required careful causal validation, later provided by the Granadillo cohort and functional studies. PMID: 31670606 shows TNRC6 is **not required for silencing by fully complementary duplexes**, clarifying that the disease-relevant function is specifically **miRNA** (partial-complementarity) regulation, not all RNAi.

Limitations and Knowledge Gaps

1. **Small evidence base:** Fewer than ~50–100 confirmed cases; phenotype frequencies rest heavily on one 17-patient cohort, limiting precision and generalizability.
2. **No patient-level molecular profiling:** No transcriptomic, proteomic, metabolomic, or DNA-methylation ("episignature") data from patients; the miRNA-derepression mechanism in human neurons is inferred, not demonstrated.
3. **No disease-specific animal model:** Existing *Tnrc6b* knockout mice are perinatal-lethal when homozygous and do not model the human heterozygous NDD; no conditional/heterozygous CNS model exists.
4. **Quantitative genetics unknown:** Penetrance and expressivity are not quantified; genotype-phenotype correlations (missense vs. truncating vs. deletion) are not established.
5. **Epidemiology absent:** No prevalence/incidence, no formal Orphanet estimate, no registry.
6. **Ontology/coding gaps:** No dedicated Orphanet/ICD/MeSH code identified; captured under broad NDD categories.
7. **No targeted therapy or trials:** Management is entirely supportive; no NCT-registered disease-specific interventions.

Proposed Follow-up Experiments / Actions

1. **Build a patient registry / GeneMatcher cohort** to expand case numbers, refine phenotype frequencies, and quantify penetrance and expressivity.

2. **Generate patient-derived iPSC neurons/organoids** carrying LoF *TNRC6B* variants and perform RNA-seq/small-RNA-seq to test for predicted miRNA-target derepression and dendritic/differentiation defects.
3. **Create a conditional/heterozygous Tnrc6b mouse (or neuronal-specific knockdown)** to establish an NDD-relevant model and test behavioral, electrophysiological, and dendritic-morphology endpoints.
4. **Define a DNA-methylation episignature** across confirmed cases to aid VUS reclassification and clinical diagnosis.
5. **Systematic genotype-phenotype analysis** separating truncating, missense, splice, and 22q13.1-deletion subgroups, including assessment of second-site modifiers (e.g., concurrent *MECP2* loss).
6. **Functional validation pipeline** (minigene/RNA assays and tethering/silencing reporters) as a standard for reclassifying splice and synonymous VUS.
7. **Natural-history study** to document developmental trajectories, seizure risk, and growth/metabolic outcomes, informing anticipatory guidance and counseling.

Report compiled from 25 reviewed papers and 9 confirmed findings across 5 investigation iterations. Evidence types span human clinical case series/reports, in vitro biochemistry and cell biology, and model-organism (mouse) genetics; mechanistic links between molecular dysfunction and clinical phenotype are explicitly flagged as inferred where direct patient-level data are lacking.
